

Project 4: Preference Elicitation

Feature representation, a Plackett–Luce extension of Bradley–Terry, and a proposed within-participant study of pairwise choice and ten-movie ranking interfaces.



PDF report for Tasks 1–3

The corresponding participant interface implements Task 4.

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Scope and summary

This project designs—but does not conduct—a study comparing two cold-start preference elicitation methods. Each participant completes 27 binary movie choices and three complete rankings of ten movies. A separate linear utility model is fitted from each method, and both models make predictions for the same 20 unseen validation pairs.

The implementation uses the supplied IMDb 5000 Movie Dataset, which contains movie metadata but no preference labels for the new participant. Task 1 defines a 53-dimensional content representation. Task 2 models a complete ranking with the Plackett–Luce sequential-choice likelihood, which reduces to Bradley–Terry for two items. Task 3 proposes a counterbalanced within-participant protocol, recruitment plan, outcomes, analysis, ethics, and validity controls. Task 4 is represented by the working Django participant interface linked to this report.

Report structure

1. Task 1 — dataset processing and movie features
2. Task 2 — Bradley–Terry and the ranking likelihood
3. Task 3 — proposed comparative user study
4. Implementation traceability and limitations

Task 1 — Feature representation

1.1 Design objective

The model assigns movie i utility $u_i = \mathbf{w}^T \mathbf{x}_i$. Therefore, $\mathbf{x}_i$ should describe attributes a person might genuinely prefer, should work for unseen movies, and should remain small enough to estimate $\mathbf{w}$ from few interactions. The representation deliberately uses content rather than aggregate audience behaviour.

1.2 Cleaning

The source CSV has 5,043 rows and 28 columns. IMDb identifiers are extracted from the URL, whitespace is normalised, numeric year and duration fields are parsed, and missing display text receives an explicit “Unknown” value. Rows are deduplicated by IMDb identifier using the first source row as a deterministic rule. This produces 4,919 unique movies. Required columns are checked and an invalid IMDb link, title, or genre raises an error instead of silently creating a malformed item.

1.3 The 53 features

Group	Count	Encoding	Reason
Genres	26	Multi-hot indicators	A movie may belong to several genres; genre is a direct and interpretable preference signal.
Release era	9	One-hot decade groups, including Unknown	Captures preference for older or newer films without treating calendar year as a strictly linear effect.
Language	7	English, French, Spanish, Hindi, Mandarin, Other, Unknown	Preserves frequent languages while avoiding unstable columns for rare categories.
Content certificate	9	Common certificates plus Other and Unknown	Represents suitability and content-intensity preferences; it is not an audience score.
Duration	2	Standardised duration and missing-value indicator	Allows preference for shorter or longer movies while preserving information that duration was absent.

Duration is median-imputed (the cleaned catalogue median is 103 minutes), clipped to the 1st and 99th percentiles, and standardised using statistics from the full catalogue. A separate binary flag marks the 15 missing durations. When a subset is transformed, the full catalogue remains the scaling reference, so the same movie has the same vector in elicitation and validation.

1.4 Excluded attributes and leakage control

IMDb score, vote count, critic reviews, Facebook likes, gross, and budget are excluded. Those fields describe crowd response, exposure, or commercial success; using them could make the model reproduce popularity rather than learn the new participant's preference. Titles and URLs are identifiers, not transferable content features. Director and actor names are shown to participants but excluded from the vector because thousands of sparse identity columns cannot be estimated reliably from only 27 likelihood contributions. Country is similarly omitted to keep the cold-start model compact; language and genre retain more direct signals.

This is a transparent baseline, not a claim that 53 hand-designed features are universally optimal. Text embeddings or adaptive feature selection could be evaluated later, but would add opacity and substantially more parameters.

1.5 Extraction method

`load_movie_catalog()` performs validation, cleaning, and deduplication. `extract_movie_features()` creates the fixed schema, and `prepare_movie_data()` returns aligned metadata and feature rows. Automated tests check shape, finiteness, category coverage, subset stability, representative values, and the absence of popularity and financial fields.

Task 2 — Model for complete rankings

2.1 Pairwise Bradley–Terry model

For participant vector $\mathbf{w}$ and movie feature vector $\mathbf{x}_i$, define $u_i = \mathbf{w}^T \mathbf{x}_i$. Bradley–Terry gives the probability that movie i is preferred to movie j as:

$$P(i > j \mid \mathbf{w}) = \exp(u_i) / [\exp(u_i) + \exp(u_j)] = \sigma(u_i - u_j).$$

2.2 Plackett–Luce extension

A complete ranking $i_1 > i_2 > \dots > i_n$ is treated as a sequence of choices. At position r , the participant selects i_r from the movies not yet placed. Multiplying these conditional choices gives:

$$P(i_1 > \dots > i_n \mid \mathbf{w}) = \prod_{r=1}^{n-1} \exp(u_{i_r}) / \sum_{s=r}^n \exp(u_{i_s}).$$

For $n = 2$, the product contains one term and is exactly the Bradley–Terry probability. The extension therefore preserves the same latent utility interpretation instead of inventing unrelated pair labels. It defines a valid probability distribution over all permutations and uses the information in the complete ordering. A ten-movie ranking contributes nine sequential choice terms; three rankings contribute 27 terms, matching the 27 binary comparisons.

2.3 Estimation

The two interfaces are fitted separately. Pairwise observations are stored as winner–loser index pairs, while ranking observations are best-to-worst index sequences. For either dataset D , the implementation minimises the negative regularised log likelihood:

$$L(\mathbf{w}) = - \sum_{y \in D} \log P(y \mid \mathbf{w}) + (\lambda/2) \|\mathbf{w}\|_2^2, \quad \lambda = 1.$$

Positive L2 regularisation is important because the feature representation is larger than the interaction budget and includes correlated indicators. The code uses analytic gradients, stable log-sum-exp computations, and L-BFGS-B. Tests verify numerical stability, gradients, fitting behaviour, and that a two-item Plackett–Luce ranking reduces to Bradley–Terry.

2.4 Assumptions

- A single linear preference vector represents a participant during the session.
- Conditional choices follow Luce’s independence-of-irrelevant-alternatives assumption.
- Displayed metadata is sufficient for the participant to form a preference.
- Errors are stochastic; the highest-utility movie need not always be chosen.

The IIA assumption is convenient and makes the model a defensible direct extension, but can be unrealistic for near-duplicate movies. A future study could compare a context-dependent ranking model or introduce random coefficients.

Task 3 — Proposed user study

3.1 Research questions and hypotheses

The study asks whether repeated pairwise choice or complete ten-movie ranking is a better way to learn a new user’s movie preferences under a similar modelling budget. The primary confirmatory hypothesis is two-sided:

- **H1:** held-out prediction agreement differs between models trained from pairwise choices and ten-movie rankings.
- **H2:** pairwise choice produces lower reported mental demand and shorter total interaction time.
- **H3:** complete ranking receives higher preference-expression ratings because it allows relative placement of more items.

H1 is two-sided because a richer ranking could provide more information, while its higher cognitive burden could also introduce noise. Directional expectations for H2 and H3 are secondary and would be declared before data collection.

3.2 Experimental design

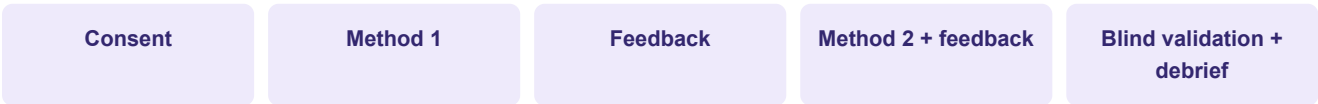
This is a within-participant crossover design: every participant uses both interfaces. This controls for between-person differences in movie knowledge and preference consistency. Order and movie-pool assignment are counterbalanced in four cells:

Cell	First condition	Second condition
1	Pairwise on pool A	Ranking on pool B
2	Ranking on pool B	Pairwise on pool A
3	Pairwise on pool B	Ranking on pool A
4	Ranking on pool A	Pairwise on pool B

The prototype maps its random seed uniformly to one of the four cells and makes each plan reproducible. In real recruitment, a concealed permuted-block schedule with one assignment to each cell per block should maintain equal cell sizes. Counterbalancing separates interface effects from order, fatigue, practice, and the particular movies assigned to a method.

3.3 Movie sampling and task budget

For each participant, 100 unique movies are sampled uniformly without replacement: 30 in pool A, 30 in pool B, and 40 in a disjoint validation pool. Pairwise tasks comprise 27 distinct pairs. All 30 pairwise-pool movies appear at least once, 24 appear twice, and six appear once; left/right orientation and task order are randomised. Ranking comprises three disjoint sets of ten, so every movie in its 30-item pool appears exactly once. Validation has 20 pairs and uses each of its 40 movies once. No validation movie is used to estimate either model.



3.4 Participants and recruitment

Recruit adults through university mailing lists and local community channels. Inclusion criteria are age 18 or older, sufficient English to understand the interface, and watching at least one movie per month. Recruitment material should describe a 15–20 minute movie-preference interface study without advertising a preferred method. Compensation should be fixed and not depend on responses.

A final sample of 80 completed participants (20 per counterbalancing cell) is a practical target. As a rough planning calculation, a paired standardised effect of 0.35 at two-sided $\alpha = .05$ and 80% power requires approximately 67 participants; the target of 80 also balances the four cells. Recruit about 88 participants to allow roughly 10% incomplete sessions. Before a real study, this approximation should be replaced by a simulation-based power analysis for the prespecified mixed-effects primary model.

3.5 Procedure

1. Show purpose, duration, voluntary-participation statement, anonymity information, age criterion, and consent checkbox.
2. Create an anonymous identifier and uniformly sample the participant's disjoint movie pools and counterbalancing cell.
3. Present the first elicitation method with clear instructions and visible progress. Hide IMDb score, votes, popularity, and financial data.
4. Immediately collect the same four 1–7 items: ease, preference expression, response confidence, and mental demand, plus an optional comment.
5. Repeat Steps 3–4 for the second method using the other movie pool.
6. Fit one regularised model from the 27 pairwise observations and another from the three rankings. Freeze both sets of 20 validation predictions.
7. Present the 20 unseen pairs without revealing either prediction and record the participant's choice and response time.
8. Show a debrief and a descriptive personal match summary. Make clear that it is not evidence that one interface is generally superior.

3.6 Measures

Type	Measure	Operational definition
Primary	Held-out agreement	Whether each frozen model predicts the participant's choice on each of 20 unseen pairs; summarised per participant.
Secondary	Probabilistic prediction	Log loss and Brier score from each model's frozen probability for the chosen movie.
Efficiency	Time	Total response time per condition and time per elementary likelihood contribution.
Experience	Four 1–7 items	Ease, preference expression, confidence, and mental demand, collected after each method.
Quality control	Completion/integrity	Complete task count, exact ranking permutations, duplicate-submission checks, and technical-error logs.

“Agreement” is used instead of “accuracy” when communicating to participants: their validation choice is a behavioural criterion, not an objectively correct label. Predictions are frozen before validation to prevent leakage.

3.7 Analysis plan

- Analyse all consenting participants who complete both methods, both questionnaires, and validation. Report attrition and reasons.
- For H1, fit a mixed-effects logistic model to validation match (yes/no), with elicitation method as the main fixed effect, participant as a random intercept, and order and pool assignment as prespecified fixed controls.
- As a participant-level robustness analysis, compare the two 20-pair match rates with a paired permutation test and bootstrap the mean paired difference with a 95% confidence interval.
- Compare log loss and Brier score as paired probabilistic outcomes. These retain confidence information that thresholded match rate discards.
- For H2, compare paired total condition times and mental-demand ratings. Inspect skew; use a paired t-test only when reasonable, otherwise a Wilcoxon signed-rank test.
- For H3 and other ordinal questionnaire items, use paired Wilcoxon tests and report matched-pairs effect sizes and confidence intervals.
- Check method-by-order interaction as a carryover diagnostic. Correct the family of secondary questionnaire tests with Holm’s method; keep H1 as the single primary test.

The analysis script and exclusion rules should be fixed before inspecting condition differences. Report estimates, uncertainty, and raw distributions, not only p-values. No imputation is planned for an incomplete primary outcome; incomplete sessions are described separately.

3.8 Exclusions, ethics, and data handling

Predefined exclusions are: under 18 or no consent, duplicate participation, failure to finish the required sequence, a documented technical failure, or implausible timing according to a threshold fixed during piloting. Exclusions must not depend on which method performs better. Sensitivity analyses should include all otherwise usable sessions.

The prototype asks for no name or email and uses an anonymous UUID. A real study requires institutional ethics review where applicable, a controller/contact and retention period in the information sheet, encrypted transport, access controls, and a persistent research database separated from identifying recruitment records. Participants may stop at any time and must still receive compensation according to the approved policy. The current Django session storage demonstrates interaction recording; production collection would export pseudonymous records under the approved data-management plan.

3.9 Validity threats and mitigations

- **Unknown movies:** metadata may not be enough to form a preference. Record uncertainty in piloting and consider a “not familiar, judge from metadata” instruction or richer neutral descriptions.
- **Cognitive burden:** ranking ten items can cause satisficing. Progress indicators, move buttons, and only three ranking tasks limit burden.
- **Order and fatigue:** the crossover is counterbalanced and order is included in analysis.
- **Movie-set difficulty:** A/B pool assignment is crossed with order, while validation is shared by both fitted models within participant.

- **Uniform sampling:** it is unbiased with respect to model beliefs but may show obscure items. Adaptive sampling is a future comparison, not mixed into this interface study.
- **Model misspecification:** linear utility and IIA may not capture context, diversity, or interactions. Conclusions concern these two interfaces under the specified model.
- **Convenience sample:** university/community recruitment limits generalisability; report participant characteristics and avoid claims about all movie users.

Implementation traceability

Requirement	Implementation
Task 1 extraction	project4/services/movie_data.py and focused data tests
Task 2 models	project4/services/preference_model.py, analytic likelihood/gradient tests
Task 3 protocol	Four counterbalancing cells, disjoint pools, balanced exposure, timing, questionnaires, frozen validation predictions
Task 4 interface	Consent, pairwise cards, reorderable ten-movie lists, feedback, validation, progress, session persistence, and debrief
Landing page	Start-study action and downloadable copy of this PDF

The implementation is a study prototype. It demonstrates the full participant sequence and computes a personal descriptive result, but no participants were recruited and no group-level experimental results are reported.

References

1. Bradley, R. A., & Terry, M. E. (1952). Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika*, 39(3/4), 324–345. <https://doi.org/10.1093/biomet/39.3-4.324>
2. Plackett, R. L. (1975). The analysis of permutations. *Applied Statistics*, 24(2), 193–202. <https://doi.org/10.2307/2346567>
3. IMDb 5000 Movie Dataset. Dataset supplied for the course project; public catalogue listing: [Kaggle dataset page](#).